

Mini-Projects: Introduction to Delphi tools

These interactive activities naturally build on each other, but can also be done in isolation if you only have a few minutes to stop by.

Let the host know if you're interested in a particular section or topic! They can set you up with all precursor code.

Follow-along with the ★ items, or choose your own path!

① symbols offer questions to help you get to know the data or system better.

✂ symbols provide “Side Quests” for exploring additional topics or diving deeper into a particular function



Access this worksheet later!

MINI-PROJECT 1: Finding indicators and fetching data

Most of this section can be done in a browser using EpiPortal. EpiPortal (delphi.cmu.edu/epiportal) catalogues both Delphi-hosted data and external data.

1.1 Explore EpiPortal and pick an indicator

Play around with the filters (pathogen, geographic level, and more) to find a Delphi-hosted indicator you're interested in. You will use this indicator in later steps.

★ **Can't decide? Choose “NHSN RSV hospital admissions”**

① What's the oldest data you can find? The most recent?

① What data sources (if any) can you find outside the United States?

✂ Side quest: Want to learn about NHSN? <https://www.cdc.gov/nhsn/about-nhsn/index.html>
<https://www.cdc.gov/nhsn/psc/hospital-respiratory-reporting.html>

1.2 Access and view data

[1] Click the grey triangle next to the indicator(s) of interest, and then **[2]** tick the checkbox next to the indicator(s) you're using. Next, **[3]** click the big blue "Show Selected Indicators" button.

1

Name ?	Pathogen(s)/ Syndrome(s)	Geographic Coverage ?	Geographic Levels ?	Temporal Scope Start ?	Temporal Scope End ?	Temporal Granularity ?	Reporting Cadence ?	Reporting Lag(nominal) ?	Revisior Cadence ?
NHSN RSV hospital admissions	RSV	United States	National U.S. HHS Region State/ADM 1	2020-08-08	Ongoing	Weekly	Weekly	4-7 days	

Total number of hospital admissions and reporting hospitals with confirmed RSV during an epi-week (Sunday-Saturday) from the National Healthcare Safety Network

2

Indicator Name	Indicator API Name	Indicator Description
☒ Confirmed RSV hospital admissions per week	confirmed_admissions_rsv_ew	RSV hospital admissions per week (t
☐ Confirmed RSV hospital admissions per week (preliminary)	confirmed_admissions_rsv_ew_prelim	RSV hospital admissions per week (t
☒ Count of hospitals reporting new RSV admissions per week	hosprep_confirmed_admissions_rsv_ew	Hospitals reporting new RSV admis
☐ Count of hospitals reporting new RSV admissions per week (preliminary)	hosprep_confirmed_admissions_rsv_ew_prelim	Hospitals reporting new RSV admis

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Name ?	Pathogen(s)/ Syndrome(s)	Geographic Coverage ?	Geographic Levels ?	T Sc
NHSN RSV hospital admissions	RSV	United States	National U.S. HHS Region State/ADM 1	202

Total number of hospital admissions and reporting hospitals with confirmed RSV during an epi-week (Sunday-Se

Indicator Name	Indic
☒ Confirmed RSV hospital admissions per week	confi
☐ Confirmed RSV hospital admissions per week (preliminary)	
☒ Count of hospitals reporting new RSV admissions per week	hoen

3

SHOW SELECTED INDICATORS

★ Select “Confirmed RSV hospital admissions per week” and “Count of hospitals reporting new RSV admissions per week”

A second view will appear:

Selected indicators

Indicator Set name

Indicator name

NHSN RSV hospital admissions

Confirmed RSV hospital admissions per week

NHSN RSV hospital admissions

Count of hospitals reporting new RSV admissions per week

Select Mode:

☒ Plot data ☐ Export data ☐ Preview Data ☐ Create query code

Location(s):

API Key (optional):

☐ Save API key (auth token) between visits

To learn more about API key, please follow this [link](#).

SUBMIT

Fill in “Location(s)” first - when you click in the box to the right of “Location(s):”, options will be displayed. Multiple options can be selected.

★ Select “United States”, “HHS Region 5”, and “Michigan”

Selected indicators

Indicator Set name

Indicator name

NHSN RSV hospital admissions

Confirmed RSV hospital admissions per week

NHSN RSV hospital admissions

Count of hospitals reporting new RSV admissions per week

Select Mode:

☒ Plot data ☐ Export data ☐ Preview Data ☐ Create query code

Location(s):

× United States

× HHS Region 5

× Michigan

API Key (optional):

☐ Save API key (auth token) between visits

To learn more about API key, please follow this [link](#).

SUBMIT

✂ Side quest: Want to learn about HHS regions?

<https://www.hhs.gov/about/agencies/iea/regional-offices/index.html> ;

https://en.wikipedia.org/wiki/United_States_Department_of_Health_and_Human_Services

1.3 Under “Select Mode:”, choose a course of action:

1.3.1 Choose “Plot data”

- To plot the data in EpiVis (delphi.cmu.edu/epivis), select "Plot data".

① Zoom into 2023 - 2026 and explore the data here - do you notice anything unexpected?

① What is the largest reported value for each geography?

1.3.2 Choose “Export data”

Selected indicators ×

Indicator Set name	Indicator name
NHSN RSV hospital admissions	Confirmed RSV hospital admissions per week
NHSN RSV hospital admissions	Count of hospitals reporting new RSV admissions per week

Select Mode: ☐ Plot data ☒ Export data ☐ Preview Data ☐ Create query code

Location(s):

✕ United States ✕ HHS Region 5 ✕ Michigan

Start Date:

01/01/2016

End Date:

01/01/2029

API Key (optional): ☐ Save API key (auth token) between visits

To learn more about API key, please follow this [link](#).

SUBMIT

- To download the data in CSV format, select "Export data" and choose the Start and End date for the data download.
 - Load and plot the data using R or Python, then continue into Mini-Project 2

1.3.3 Choose “Create query code”

If you don't already have Delphi packages set up on your computer, please see the host for access to an already-configured environment.

- To access data through the Delphi Epidata API, click "Create query code". We will use `epidatr` ([documentation](#)) for now, but a Python client, `epidatpy` ([documentation](#)), is also available.

Selected indicators ×

Indicator Set name	Indicator name
NHSN RSV hospital admissions	Confirmed RSV hospital admissions per week
NHSN RSV hospital admissions	Count of hospitals reporting new RSV admissions per week

Select Mode: ☐ Plot data ☐ Export data ☐ Preview Data ☒ Create query code

Location(s):

× United States × HHS Region 5 × Michigan

Start Date:

End Date:

API Key (optional):

☐ Save API key (auth token) between visits

To learn more about API key, please follow this [link](#).

SUBMIT

- Select “Submit” and then scroll down to the R code section (it’s below the Python code)
 - Copy the scripts into R
- Convert the data into `epi_df` format using `epiprocess::as_epi_df`.

① Use the `class()` function to identify what has changed (if anything) about the object classification of your data.

- Plot the data using `epiprocess::autoplot`. [Hint: To view the help file for this method, type `?autoplot.epi_df` in R or visit [this link](#).]

① What do you notice about the data once it's plotted? What time range does it cover? Are there any patterns of note?

- Then continue into Mini-Project 2

Example solutions!

